

Topic to discuss

→ Lagrange's Interpolation

→ 2 Numerical Problem

→ Homework Problem with solution.

YouTube @StartPracticing

Interpolation for unequal Interval

Interpolation for unequal intervals means x -values are not equally spaced, so we can't directly use simple Newton forward or backward difference formula. In this case, methods like Lagranges or Newton's divided difference are used because they work for any spacing between points.

Lagrange's Interpolation

Lagrange's interpolation is a numerical method used to find a single polynomial that passes exactly through a given set of data points.

Suppose we know the values like $(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$ and we want to estimate the function value at some new point x (between or near them). Lagrange's interpolation build a polynomial $P_n(x)$ of degree at most n such that:

$$P_n(x_i) = y_i \quad \forall i = 0, 1, \dots, n$$

Polynomial

General Form :-

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$$

- $a_0, a_1, a_2 \dots a_n \rightarrow$ Coefficients (real numbers)
- $x \rightarrow$ variable
- $n \rightarrow$ degree of polynomial (highest power of x)

Example of polynomial :

- 5 $\rightarrow$ Constant
- $2x+3 \rightarrow$ Linear
- $x^2+4x+1 \rightarrow$ Quadratic
- $x^3-2x+7 \rightarrow$ Cubic

Not polynomial (invalid case) :

- $x^{-1}+2 \rightarrow$ negative exponent
- $\sqrt{x}+1 \rightarrow$ fractional exponent
- $\frac{1}{x}+4 \rightarrow$ Variable in denominator
- $2^x+1 \rightarrow$ Variable in exponent

Degree of Polynomial :

- Degree = highest power of the variable
- for eg, $4x^5+x^2+3$ has degree 5

Lagrange's Interpolation Formula

Considered the data points $(x_0, y_0), (x_1, y_1), (x_2, y_2), (x_3, y_3)$ and (x_4, y_4) with unequal interval, then by Lagrange's formula $f(x)$ or $y(x)$ is given by :

$$\begin{aligned} y(x) = & \frac{(x-x_1)(x-x_2)(x-x_3)(x-x_4)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)(x_0-x_4)} y_0 \\ & + \frac{(x-x_0)(x-x_2)(x-x_3)(x-x_4)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)(x_1-x_4)} y_1 \\ & + \frac{(x-x_0)(x-x_1)(x-x_3)(x-x_4)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)(x_2-x_4)} y_2 \\ & + \frac{(x-x_0)(x-x_1)(x-x_2)(x-x_4)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)(x_3-x_4)} y_3 \\ & + \frac{(x-x_0)(x-x_1)(x-x_2)(x-x_3)}{(x_4-x_0)(x_4-x_1)(x_4-x_2)(x_4-x_3)} y_4 \end{aligned}$$

Numerical Problem - 1 :

Q:- Find the value of Y at $X=21$ using Lagrange's Interpolation polynomial from the following table:

X	20	23	26	29
Y	0.3420	0.3907	0.4384	0.4848

Solution : Given,

$$x_0 = 20, x_1 = 23, x_2 = 26, x_3 = 29$$

$$\text{and } y_0 = 0.3420, y_1 = 0.3907, y_2 = 0.4384, y_3 = 0.4848$$

$$\text{also, } x = 21.$$

Applying Lagrange's Interpolation formula,

$$y(x) = \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)} y_0 + \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)} y_1$$

$$+ \frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)} y_2 + \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)} y_3$$

	x_0	x_1	x_2	x_3
x	20	30	26	29
y	0.3420	0.3907	0.4384	0.4848

$y_0 \quad y_1 \quad y_2 \quad y_3$

$x = 21$

$$= \frac{(21-23)(21-26)(21-29)}{(20-23)(20-26)(20-29)} \times 0.3420 + \frac{(21-20)(21-26)(21-29)}{(23-20)(23-26)(23-29)} \times 0.3907$$

$$+ \frac{(21-20)(21-23)(21-29)}{(26-20)(26-23)(26-29)} \times 0.4384 + \frac{(21-20)(21-23)(21-26)}{(29-20)(29-23)(29-26)} \times 0.4848$$

$$= \frac{-2 \times -5 \times -8}{-3 \times -6 \times -9} \times 0.3420 + \frac{1 \times -5 \times -8}{3 \times -3 \times -6} \times 0.3907$$

$$+ \frac{1 \times -2 \times -8}{6 \times 3 \times -3} \times 0.4384 + \frac{1 \times -2 \times -5}{9 \times 6 \times 3} \times 0.4848$$

$$= 0.1688 + 0.2894 - 0.12989 + 0.02992$$

$$= 0.35823$$

Ans

Numerical Problem - 2:

Given the following data, find $\log_{10} 656$.

x	654	658	659
$\log_{10} x$	2.8156	2.8182	2.8189

Solution :-

Given, $x_0 = 654$, $x_1 = 658$, $x_2 = 659$

and $y_0 = 2.8156$, $y_1 = 2.8182$, $y_2 = 2.8189$

and $x = 656$.

Applying Lagrange's Interpolation formula,

$$y(x) = \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)} y_0 + \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)} y_1 + \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)} y_2$$

	x_0	x_1	x_2
x	654	658	659
$\log_{10} x$	2.8156	2.8182	2.8189

y_0 y_1 y_2

$$x = 656$$

$$= \frac{(656-658)(656-659)}{(654-658)(654-659)} \times 2.8156 + \frac{(656-654)(656-659)}{(658-654)(658-659)} \times 2.8182 + \frac{(656-654)(656-658)}{(659-654)(659-658)} \times 2.8189$$

$$= \frac{-2x-3}{-4x-5} \times 2.8156 + \frac{2x-3}{4x-1} \times 2.8182 + \frac{2x-2}{5x-1} \times 2.8189$$

$$= 0.84468 + 4.2273 - 2.25512$$

$$= 2.81686 \text{ Ans.}$$

Homework Problem - 1 :

Q: Find the interpolation polynomial for the function $y = f(x)$ given by the table:

x	-1	1	4	6
y	1	-3	21	127

Solution : To construct the interpolation polynomial, we have to apply Lagrange's interpolation formula,

$$x_0 = -1, \quad y_0 = 1$$

$$x_1 = 1, \quad y_1 = -3$$

$$x_2 = 4, \quad y_2 = 21$$

$$x_3 = 6, \quad y_3 = 127$$

$$f(x) = \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)} y_0 + \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)} y_1$$

$$+ \frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)} y_2 + \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)} y_3$$

$$= \frac{(x-1)(x-4)(x-6)}{(-1-1)(-1-4)(-1-6)} \times 1 + \frac{(x+1)(x-4)(x-6)}{(1+1)(1-4)(1-6)} \times -3$$

$$+ \frac{(x+1)(x-1)(x-6)}{(4+1)(4-1)(4-6)} \times 21 + \frac{(x+1)(x-1)(x-4)}{(6+1)(6-1)(6-4)} \times 127$$

$$= \frac{(x-1)(x^2-10x+24)}{-70} + \frac{(x+1)(x^2-10x+24)}{30} \times -3 + \frac{(x+1)(x^2-7x+6)}{-30} \times 21 + \frac{(x^2-1)(x-4)}{70} \times 127$$

$$= -\frac{1}{70} (x^3 - 11x^2 + 34x - 24) - \frac{1}{10} (x^3 - 9x^2 + 14x + 24) - \frac{7}{10} (x^3 - 6x^2 - x + 6) + \frac{127}{70} (x^3 - 4x^2 - x + 4)$$

$$= x^3 \left(-\frac{1}{70} - \frac{1}{10} - \frac{7}{10} + \frac{127}{70} \right) + x^2 \left(\frac{11}{70} + \frac{9}{10} + \frac{42}{10} - \frac{4 \times 127}{70} \right) + x \left(-\frac{34}{70} - \frac{14}{10} + \frac{7}{10} - \frac{127}{70} \right) + \left(\frac{24}{70} - \frac{24}{10} - \frac{42}{10} + \frac{4 \times 127}{70} \right)$$

$$= x^3 \left(\frac{-1-7-49+127}{70} \right) + x^2 \left(\frac{11+63+294-508}{70} \right) + x \left(\frac{-34-98+49-127}{70} \right) + \left(\frac{24-168-294+508}{70} \right)$$

$$= x^3 - 2x^2 - 3x + 1$$

Ans.

Connect with me



[Start Practicing](#)



[i._am._arfin](#)



[Arfin Parween](#)



[Arfin Parween \(\[Subscribe here too\]\(#\) \)](#)



[Arfin Parween](#)



[Arfin Parween](#)